

Entropy-Maximized Multiplicative Weight Noise for Subject-Disjoint Robustness

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Abstract

We implemented an EntropyMaxWeightNoise module within a PreNormTransformerEncoder and evaluated performance on the UCI Human Activity Recognition Using Smartphones dataset. The study investigates whether explicit entropy maximization of weight uncertainty parameters during training improves out-of-distribution accuracy on subject-disjoint time-series data compared to standard deterministic training. We trained models for 12 epochs using SGD across 43 seeds in a primary cohort and 43 seeds in a confirmatory replication cohort. The hypothesis predicted that the intervention would yield higher held-out classification accuracy than the comparator, with a mean hypothesis-positive paired contrast exceeding 0.0125098. In the primary cohort, the intervention worsened the contrast by 0.04805 (95% CI [-0.0714, -0.0324], $p=0.0002$). In the replication cohort, the intervention worsened the contrast by 0.03482 (95% CI [-0.0422, -0.0278], $p=0.0002$). Both cohorts resulted in a preregistered decision of `direction_refuted`. These findings suggest that entropy maximization of multiplicative weight noise does not generalize to attention-based architectures under distribution shift in this setting.

1. Introduction

Confidence in neural network predictions is increasingly important, yet basic neural networks do not deliver certainty estimates [3]. Modern deep learning methods operate as black boxes, making the uncertainty associated with their predictions challenging to quantify [12]. Bayesian statistics offer a formalism to understand and quantify this uncertainty [12]. Neural networks intrinsically increase uncertainty about which features of the analysis are model-related and which are due to the neural network [5]. Uncertainty evaluation is a core technique when deep neural networks are used in real-world problems [4]. However, standard backpropagation has disadvantages, such as a lack of calibrated probabilistic predictions [17].

Bayesian neural networks (BNNs) promise improved generalization under covariate shift [6]. However, weight-based BNNs often struggle with high computational complexity [6]. Employment of BNNs is still constrained by increased computational requirements and convergence difficulties [7]. Bayesian neural network posterior distributions have a great number of modes that correspond to the same network function [18]. Approximate Bayesian inference for neural networks is considered a robust alternative to standard training, often providing good performance on out-of-distribution data [13]. However, BNNs with high-fidelity approximate inference can achieve poor generalization under covariate shift [13].

This study addresses whether uncertainty-inducing mechanisms optimized for diversity (entropy maximization) rather than posterior fidelity (KL minimization) generalize to attention-based architectures under distribution shift. While prior work explores weight noise injection and Bayesian uncertainty separately, no study explicitly maximizes the entropy of multiplicative weight noise distributions to improve subject-disjoint generalization in Transformers against standard variational objectives that minimize divergence. We test the hypothesis that explicit entropy maximization of weight uncertainty parameters during training improves out-of-distribution accuracy on subject-disjoint time-series data compared to standard deterministic training.

2. Related Work

Research into uncertainty quantification establishes that confidence in neural network predictions is increasingly important, yet **A survey of uncertainty in deep neural networks** notes that basic neural networks do not deliver this confidence [3]. **Hands-On Bayesian Neural Networks—A Tutorial for Deep Learning Users** describes deep learning methods as black boxes where uncertainty is challenging to quantify, stating that Bayesian statistics offer a formalism to address this [12]. **Bayesian Neural Networks** observes that neural networks intrinsically increase uncertainty about which features of the analysis are model-related and which are due to the neural network [5]. **Uncertainty propagation for dropout-based Bayesian neural networks** emphasizes uncertainty evaluation as core for detecting uncertain data in practical applications [4], while **Probabilistic Backpropagation for Scalable Learning of Bayesian Neural Networks** highlights disadvantages of standard backpropagation such as lack of calibration [17]. However, **Stochastic Weight Sharing for Bayesian Neural Networks** and **Structured Partial Stochasticity in Bayesian Neural Networks** identify constraints regarding computational requirements, convergence difficulties, and posterior modes in BNNs [7, 18], and **Bayesian Neural Networks for Macroeconomic Analysis** develops BNNs specifically for data with limited observations [16].

Regarding robustness and optimization, **Tackling covariate shift with node-based Bayesian neural networks** states BNNs promise improved generalization under covariate shift but weight-based BNNs often struggle with high computational complexity [6]. **Dangers of Bayesian Model Averaging under Covariate Shift** warns that approximate Bayesian inference considered robust can achieve poor performance under covariate shift [13], while **Pretrained Transformers Improve Out-of-Distribution Robustness** systematically measures out-of-distribution generalization for pretrained Transformers on NLP datasets [8]. **Batch Normalization: Accelerating Deep Network Training by Reducing Internal Covariate Shift** addresses the change in distribution of layer inputs during training [9], and **A Diffractive Neural Network with Weight-Noise-Injection Training** proposes weight noise injection training to achieve strong robustness [15]. **Alignment Entropy Regularization** uses entropy to measure a model's uncertainty in distributing probability mass over sequences [14], and **Bayesian optimization with robust Bayesian neural networks** discusses Bayesian optimization for tuning hyperparameters of machine learning algorithms [11]. Additionally, **Precision-Dependent Breakdown of Momentum in Ill-Conditioned Linear Regression** and **Stdlib Gradient Descent Learning Rate Sweep on Synthetic Non-Linear Data** investigate convergence stability and learning rate effects in optimization [1, 2], while **1D convolutional neural networks and applications: A survey** describes CNNs as standard for various operations [10].

3. Method

We implemented EntropyMaxWeightNoise as a custom PyTorch module wrapping the query projection layer. The model architecture was a PreNormTransformerEncoder consisting of a Linear(9,64) input layer, a learned positional embedding(128,64), and 2 blocks. Each block contained LayerNorm(64), 4-head self-attention over 128 timesteps with q/k/v/out Linear(64,64), a residual connection, LayerNorm(64), Linear(64,128), ReLU, Linear(128,64), and a final residual connection. The head consisted of a final LayerNorm(64), mean-pool over time, and Linear(64,6).

We evaluated performance on the UCI Human Activity Recognition Using Smartphones dataset (DOI 10.24432/C54S4K), comprising raw inertial signals in 128-timestep windows over 9 channels. We used the official subject-disjoint split of 7,352 training and 2,947 held-out windows. We ran 4 conditions: (1) Intervention (EntropyMax), (2) Baseline (Deterministic), (3) NegCtrl1 (EntropyMin), and (4) NegCtrl2 (FixedNoise). We trained models for 12 epochs with SGD. For each seed, we initialized identical model weights. We evaluated on held-out subjects and aggregated accuracy per seed. We computed paired contrasts (Intervention - Baseline).

4. Experimental Setup

We applied the preregistered decision rule on the Primary block, then confirmed on the Replication block. The Primary cohort consisted of 43 seeds. The Confirmatory replication cohort consisted of

43 seeds. We reported each metric's mean with a bootstrap 95% confidence interval. We computed paired randomisation p-values and Cohen's d_z effect sizes. The hypothesis predicted that the intervention would yield higher held-out classification accuracy than the comparator, with mean hypothesis-positive paired contrast H_s exceeding the fixed worthwhile margin of 0.0125098 proportion.

Executed configuration (server-recorded): the reported run executed the calibrated testbed ``uci_har_small_transformer_v1`` exactly as registered:

- Data: UCI Human Activity Recognition Using Smartphones (DOI 10.24432/C54S4K; raw inertial signals, 128-timestep windows over 9 channels; official subject-disjoint split of 7,352 training and 2,947 held-out windows). This is real recorded data, not synthetic or simulated: it was downloaded, checksum-verified and staged read-only to the compute node before the job started.
- ``uci_har_smartphones`` — <https://archive.ics.uci.edu/static/public/240/human+activity+recognition+using+smartphones.zip> (SHA-256 c00b803081a5c797cd5e4b83700a9810b38d53d9d84e01917e090e1fdb81031, 61005872 bytes, CC BY 4.0)
- Preprocessing: standardise each of the 9 inertial channels by subtracting that channel's mean and dividing by its standard deviation (floored at $1e-6$) computed over the TRAINING split only, so no held-out statistic reaches training.
- Model: PreNormTransformerEncoder(Linear(9,64) + learned positional embedding(128,64); 2 blocks, each LayerNorm(64) -> 4-head self-attention over 128 timesteps with q/k/v/out Linear(64,64) -> residual -> LayerNorm(64) -> Linear(64,128) -> ReLU -> Linear(128,64) -> residual; final LayerNorm(64), mean-pool over time, Linear(64,6)).
- Training budget: 12 epochs; batch size 128; SGD(lr=0.05,momentum=0.9).
- Held-out evaluation: the official UCI HAR test split — 2,947 windows recorded from 9 subjects who appear in no training window, so the held-out cohort is subject-disjoint rather than a random row split, and is never used for training or tuning.
- Outcome: held-out classification accuracy — number of correct activity predictions / 2947 windows in the official subject-disjoint UCI HAR test split (proportion).

5. Results

****Primary Cohort****

Across 4 conditions and 43 seeds, we report each metric's mean with a bootstrap 95% confidence interval (Figure 1). On 'held-out classification accuracy', 'proposed' worsened the hypothesis-positive contrast by 0.04805 versus 'baseline_1' (95% CI [-0.0714, -0.0324], paired randomisation $p=0.0002$, Cohen's $d_z=-0.72$). This is a statistically significant difference as the 95% CI excludes zero. The mean accuracy for 'proposed' was 0.8604 [0.837, 0.877], while 'baseline_1' was 0.9085 [0.905, 0.912]. The negative controls performed similarly to the baseline: 'negative_control_1' was 0.9082 [0.905, 0.911] and 'negative_control_2' was 0.909 [0.906, 0.912]. Training dynamics are shown in Figure 2. The preregistered decision is `direction_refuted`.

****Confirmatory Replication Cohort****

Across 4 conditions and 43 seeds, we report each metric's mean with a bootstrap 95% confidence interval (Figure 3). On 'held-out classification accuracy', 'proposed' worsened the hypothesis-positive contrast by 0.03482 versus 'baseline_1' (95% CI [-0.0422, -0.0278], paired randomisation $p=0.0002$, Cohen's $d_z=-1.41$). This is a statistically significant difference as the 95% CI excludes zero. The mean accuracy for 'proposed' was 0.8717 [0.865, 0.877], while 'baseline_1' was 0.9065 [0.903, 0.91]. The negative controls performed similarly to the baseline: 'negative_control_1' was 0.908 [0.904, 0.912] and 'negative_control_2' was 0.9088 [0.905, 0.912]. Training dynamics are shown in Figure 4. The preregistered decision is `direction_refuted`.

Registered-report decision: the primary cohort was classified as ****preregistered direction refuted**** and the independent confirmatory replication was classified as ****preregistered direction refuted**** under the preregistered confidence-interval rule. These cohorts were not pooled, and this outcome is reported regardless of direction.

6. Discussion

The results refute the hypothesis that explicit entropy maximization of weight uncertainty parameters improves out-of-distribution accuracy on subject-disjoint time-series data compared to standard deterministic training. In both the primary and replication cohorts, the intervention significantly worsened held-out classification accuracy relative to the baseline. This finding contradicts the expectation that diversity-optimized uncertainty mechanisms would generalize better under distribution shift.

These results relate directly to prior work on Bayesian neural networks and robustness. *Tackling covariate shift with node-based Bayesian neural networks* states BNNs promise improved generalization under covariate shift [6]. However, our findings align with *Dangers of Bayesian Model Averaging under Covariate Shift*, which warns that approximate Bayesian inference considered robust can achieve poor performance under covariate shift [13]. The observed degradation suggests that maximizing entropy of multiplicative weight noise may introduce excessive instability or over-regularization in Transformer architectures, unlike the node-based approaches described in [6].

Furthermore, *A Diffractive Neural Network with Weight-Noise-Injection Training* proposes weight noise injection training to achieve strong robustness [15]. Our negative controls (FixedNoise) performed similarly to the deterministic baseline, indicating that noise injection alone did not harm performance, but the entropy maximization objective specifically drove the degradation. This distinguishes the mechanism from simple weight noise injection [15]. *Alignment Entropy Regularization* uses entropy to measure a model's uncertainty in distributing probability mass over sequences [14]. Our results suggest that applying entropy maximization to weight uncertainty parameters in Transformers does not yield the same benefits as sequence alignment entropy [14].

The consistency between the primary and replication cohorts strengthens the conclusion that this intervention is not beneficial for this task. The effect sizes (Cohen's d_z = -0.72 in Primary, -1.41 in Replication) indicate a substantial negative impact. Practitioners should not prioritize entropy maximization for robustness tasks over standard regularization in this setting.

7. Limitations

This study is limited to the UCI Human Activity Recognition Using Smartphones dataset. While the subject-disjoint split provides a valid test of distribution shift, results may not generalize to other time-series domains or modalities. The architecture was a specific PreNormTransformerEncoder; effects may differ in deeper Transformers or CNNs. The training budget was fixed at 12 epochs; longer training might alter the convergence of the entropy maximization objective. Finally, we evaluated only held-out classification accuracy; other uncertainty metrics (e.g., calibration) were not assessed.

Declared scope. This is a preregistered, single-testbed registered report: every claim in this paper is scoped to the calibrated testbed uci_har_small_transformer_v1 and to the preregistered smallest worthwhile effect of 0.012509797746985206 proportion on held-out classification accuracy, a margin derived from the calibration's measurement noise (it reflects measurement precision, not scientific importance). Generalisation beyond this testbed and margin is explicitly out of scope and untested here.

8. Conclusion

We evaluated whether explicit entropy maximization of weight uncertainty parameters improves out-of-distribution accuracy on subject-disjoint time-series data. The hypothesis predicted higher held-out classification accuracy for the intervention. In the primary cohort, the intervention worsened accuracy by 0.04805 (95% CI [-0.0714, -0.0324]). In the replication cohort, the intervention worsened accuracy by 0.03482 (95% CI [-0.0422, -0.0278]). Both cohorts yielded a preregistered decision of direction_refuted. These findings indicate that entropy maximization of

multiplicative weight noise does not improve robustness in this context and may significantly degrade performance compared to deterministic training.

Reproducibility

All results were produced by an automated pipeline running the experiment on GWDG HPC inside a container. This study was preregistered before the experiment ran: the hypothesis, the predicted direction, the metrics, and the analysis plan (statistics + seeds) were fixed in advance and committed to the artifact repository (PREREGISTRATION.md). Random seeds:

[497360171578835126, 619747419151856849, 4359311500859520721, 3900812323465083926, 1516445550462428226, 1630043136886007277, 1835401958502649079, 4224212089476219622, 2975544019134720152, 3507764674343915068, 4004663391355442268, 1509823930515868492, 3483967100743641674, 2225936250788509260, 1957754086644492888, 4154286735266072875, 3813494016017719200, 690472406820790188, 3813916678459859089, 4090465763900842853, 1844212296145829516, 3081923774287884327, 3477292413922411937, 726925812904865219, 4400454224498690158, 1309720931756809616, 3670000523134785215, 1572470746349086798, 5569990145947559, 2855297384752490100, 3166602034212987559, 4221459448664290905, 2934143650096726109, 1462463051419223747, 4363375170763905339, 3648570110071838823, 258115887333994485, 2427086170940110629, 4084426303853737760, 2996901096067535381, 2815367292871095378, 4593572677630054665, 3754669615394449950]. Slurm job id: 15540577. Container image: pytorch/pytorch@sha256:c8268a92a69bd500f8be0e665b2630ee006dadaf7bfbcb24249141b15ff622755. Walltime: 01:30:00. The exact code, container reference, seeds, run logs, Slurm job id, and the agent's full reasoning trace (including failed attempts) are in the artifact repository: (pending publication).

Ethics

This is a small-scale computational study using synthetic or small public data and no human subjects or personal data; it poses no foreseeable ethical or dual-use risks. Compute use was deliberately minimal.

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